Physics Solve

1. State Gauss's theorem and apply it to obtain an expression for the electric field at a point. i) outside and ii) inside a charged conducting sphere.

Ans: i) At point outside the solid sphere ($r > R$)

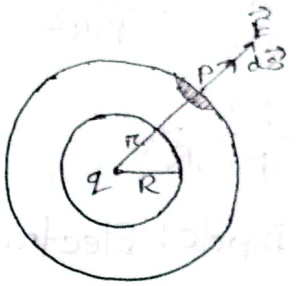
$$\oint_S \vec{E} d\vec{s} = \frac{q}{\varepsilon_0}$$

$$\Rightarrow \oint_S E ds \cos 0^\circ = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E \oint_S ds = \frac{q}{\varepsilon_0}$$

$$\Rightarrow E \times 4\pi r^2 = \frac{q}{\varepsilon_0}$$

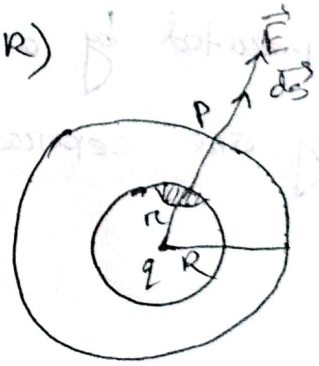
$$\Rightarrow E = \frac{q}{4\pi\varepsilon_0 r^2}$$



ii) At point inside the solid sphere ($r < R$)

ρ = volume charge density

$$\rho = \frac{q}{4/3 \pi R^3}$$



q_{in} = ρ × volume of gaussian surface

$$= \frac{q}{4/3 \pi R^3} \times 4/3 \pi r^3$$

$$= \frac{q r^3}{R^3}$$

According to gauss's lawe we can write

$$\oint_S \vec{E}\,d\vec{s} = \frac{q_{in}}{\epsilon_0}$$

$$\Rightarrow \oint_S E\,ds\cos 0^\circ = \frac{q_{in}}{\epsilon_0}$$

$$\Rightarrow E\oint_S ds = \frac{q_{in}}{\epsilon_0}$$

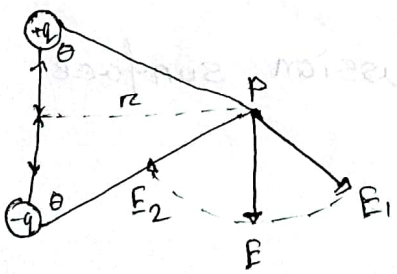
$$\Rightarrow E \times 4\pi r^2 = \frac{qr^3}{R^3\epsilon_0}$$

$$\Rightarrow E = \frac{qr^3}{R^3\epsilon_0} \times \frac{1}{4\pi r^2}$$

$$= \frac{qr}{4\pi\epsilon_0 R^3}$$

b) Electric Dipole: Electric dipole is a pair of equal and opposite charges $+q$ and $-q$ separated by a small distance $2a$.

An electric dipde consist of two equal but opposite charges seperated by a distance in figure the charge $+q$ and $-q$. They are seperated by a distance $2a$. we shall assume $x \gg a$.



Let the distance p be y from both the charges. Let E_1 be the electric field at the point p due to the charges

solve

$+q$ and E_2 be the due to the $-q$.

The total field at P due to the both dipole obtained by vector addition of

$$E = E_1 + E_2$$

The magnitudes of the fields E_1 and E_2 are equal.

$$E_1 = E_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{y_2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{(a^2+x^2)}$$

The vector sum of E_1 and E_2 points vertically downwards and has the magnitude.

$$\cos\theta = \frac{a}{y} = \frac{a}{\sqrt{a^2+x^2}}$$

$$E = \frac{2}{4\pi\epsilon_0} \frac{q}{(a^2+x^2)} \cdot \frac{a}{\sqrt{a^2+x^2}}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{2aq}{(a^2+x^2)^{3/2}}$$

Since $x > 7a$, a^2 can be neglected in denomination. Then the above equation is.

$$E = \frac{1}{4\pi\epsilon_0} \frac{2aq}{x^3}$$

c) Two parallel plates of area 200 m^2 each are seperated by a distance of 10mm. If one plate carries a charge 2×10^{-9} C and the other carries a charge -2×10^{-9} C calculate the electric field in the space between them.

d) given,

Area $A = 20cm^2$

charge $q = 2 \times 10^{-9} C$

$$\sigma = \frac{q}{A} = \frac{2 \times 10^{-9}}{20 \times 10^{-4}} cm^{-2}$$

$$= 10^{-6} cm^{-2}$$

since one of the plate carries positive charge while the other carries negative charge. The electric field in the space between them

$$E = \frac{\sigma}{\varepsilon_0} = \frac{10^{-6}}{8.85 \times 10^{-12} c^2 N^{-1} m^{-2}}$$

$$= 1.13 \times 10^5 N/c$$

2. a) Define i) susceptibility ii) permeability iii) self inductance.

Ans:

susceptibility: susceptibility is a property of material that describes its response to an external influence such as an external electric or magnetic field. in quantifies how easy the material can become polarize or magnetized when subjected to such an influence

permeability: permeability is the ability of a material to allow the magnetic flux where object is placed inside

the magnetic field where magnetic flux is measure to the number of magnetic line of forced that can pass via a given surface. permiability is denoted by a greek word 'm', 'm' measured in farad/meter (f/m)

Self inductance:- self inductance is the property of the current carrying coil that resists or opposses the change of current flowing through it.

b)

Let us assume that the secondary coil is an open circuit.

Now, $\mathcal{E}_s = N_s \frac{\Delta \phi_s}{\Delta t}$ also $\mathcal{E}_s = M \frac{\Delta i_p}{\Delta t}$

From the two eqn we get,

$M = N_s \frac{\Delta \phi_s}{\Delta i_p}$ Now $\Delta \phi_s = (2.5 \times 10^{-4} - 0)$ wb

$\Delta i_p = (2 - 0)$ A

$\therefore M = N_s \frac{(2.5 \times 10^{-4} - 0)\text{ wb}}{(2-0)\text{ A}}$

$= (1.25 \times 10^{-4}\ N_s)$ H

c) Gauss law in dielectric medium:

Let us take a parallel plate capacitor if there is no dielectric between the plate then the Gauss law.

$$\Rightarrow \varepsilon_0 \oint \vec{E}_0 \cdot d\vec{s} = q$$

$$\Rightarrow \varepsilon_0 E_0 \oint ds = q$$

$$\Rightarrow \varepsilon_0 E_0 A = q$$

$$\Rightarrow E_0 = \frac{q}{\varepsilon_0 A} \quad \text{——(I)}$$

If dielectric present then,

$$\varepsilon_0 \oint \vec{E} \cdot d\vec{s} = q - q'$$

$$\Rightarrow \varepsilon_0 E \oint ds = q - q'$$

$$\Rightarrow \varepsilon_0 E A = q - q'$$

$$\Rightarrow E = \frac{q}{\varepsilon_0 A} - \frac{q'}{\varepsilon_0 A} \quad \text{——(II)}$$

From the definition of dielectric constant K

$$K = \frac{E_0}{E}$$

$$E = \frac{E_0}{K} = \frac{\frac{q}{\varepsilon_0 A}}{K} = \frac{q}{\varepsilon_0 K A} \quad \text{——(III)}$$

Comparing eqn (II) and (III)

$$\therefore \frac{q}{\varepsilon_0 K A} = \frac{q}{\varepsilon_0 A} - \frac{q'}{\varepsilon_0 A}$$

$$\Rightarrow \frac{q}{K} = q - q'$$

so, the Gauss law,

$\epsilon_0 \oint \vec{E} d\vec{s} = q/k$

$\Rightarrow k\epsilon_0 \oint \vec{E} d\vec{s} = q$

This is Gauss law in dielectric medium.

3. a)

Current density: The current per unit cross sectional area of a conductor is called current density. It is denoted as j.

$$j = \frac{dI}{da} \quad A/m^2$$

$$\Rightarrow I = \int \vec{j} . da$$

$$\Rightarrow I = jA$$

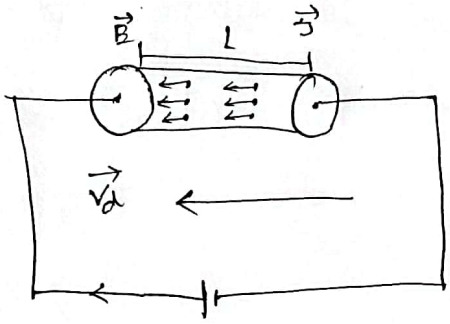
$$\Rightarrow j = \frac{I}{A}$$

The direction of current density $\vec{j}$ and applied electric field $\vec{E}$ is same but direction of drift velocity v_d of electrons is opposite to direction of current density.

Drift velocity: The free electrons move from low potential terminal to high potential terminal of battery when a conductor is connected with a battery. The kinetic energy of accelarating electrons is converted into

vibrational energy of the lattice.

The constant average velocity acquired during this process by free electrons against electric field is called drift velocity V_d. The drift velocity is of the order 10^{-3} m/s. The random thermal motion of conduction electrons in a metal takes place with typical speeds of 10^6 m/s



Let, The conductor's length = L

cross sectional Area = A

Volume of conductor = AL

Total charge carriers,

unit volume of conductor has charge carriers = n

AL " " " " " " " = nAL

total charge carriers in conducton = nAL

Now, cosider a single charge carries in conducton

one charge carrier has charge = e

∴ nAL " " " = ALne

Total charge q on conductor is

$$q = nALe \quad \text{——} (I)$$

The current established in conductor when charge q flows across the conductor in time t is.

$$I = \frac{q}{t}$$

$$\Rightarrow q = It \quad \text{——} (II)$$

compare (I) and (II)

$$It = nALe$$

The changes moves with drift velocity v_d across the conductor having length L in time to so $L = v_d t$

$$It = nA(v_d t)e$$

$$\Rightarrow I = nAv_d e$$

$$\Rightarrow \frac{I}{A} = nAv_d e$$

$$\Rightarrow \frac{I}{A} = nev_d$$

$$\Rightarrow \vec{J} = nev_d$$

The direction of drift velocity and current density is opposite to each other. so in vector from we can write.

$$\vec{J} = -nev_d$$

b) Energy stored in a capacitor

The energy of a charged capacitor is the amount of work done in charging it Let V. is the potential difference between the plate of the capacitor of capacitance C.

The work done is given by.

$$dw = VdQ$$
$$dw = Q/C\, dQ$$

now total work done

$$w = \int_{Q=0}^{Q=Q} Q/C\, dQ$$

$$w = \frac{1}{2}\frac{Q^2}{C}$$

$$= \frac{1}{2} CV^2$$

$$= \frac{1}{2} QV$$

for a parallel plate capacitor the C is given by,

$$C = \frac{\varepsilon_0 A}{d}$$

$$V = Ed$$

The energy stored as potential energy is given by

$$U = \frac{1}{2}\frac{\varepsilon_0 A}{d} E^2 d^2$$

$$= \frac{1}{2} \varepsilon_0 A E^2 d$$

so energy stored per unit volume $= \frac{1}{2}\epsilon_0 E^2$

(c) Soln:

$$j = \frac{I}{A}$$

$$= \frac{5.0\ A}{3.0 \times 10^{-6}\ m^2}$$

$$\Rightarrow 1.67 \times 10^6\ A/m^2$$

Given.
$A = 3.0\ mm^2$
$= 3.0 \times 10^{-6}\ m^2$
$I = 5.0\ A$

drift velocity is given by,

$$V_d = \frac{J}{ne} = \frac{1.67 \times 10^6\ A/m^2}{n(1.60 \times 10^{-19} C)} = \frac{1}{n}(1.04 \times 10^{25}\ m^{-2}.s^{-1})$$

where n is the member of charge carriers per unit volume of copper.

Assuming one free electron per atom of copper.

$$n = \frac{(\text{Avogardo's number})(\text{density of copper})}{\text{molecular weight of copper}}$$

$$= \frac{(6.02 \times 10^{26}\ \text{atoms/kmol})(8920\ kg/m^3)}{63.5\ kg/kmol}$$

$$= 8.5 \times 10^{28}\ \text{atoms}/m^3$$

$$V_0 = \frac{1.04 \times 10^{25}\ m^{-2}s^{-1}}{8.5 \times 10^{28}\ \text{atoms}/m^3}$$

$$= 0.122 \times 10^{-3}\ m/s$$

$$= 0.122\ mm/s$$

1.a) state and explain kinchoff's law.

Ans:

At any junction the sum of the current entering the junction is equal to the sum of currents leaving the junction.

and the algebric sum of the products of the current and resistance of each part of the circuit is equal to the total emf's acting in the circuit.

1st law - current law

$\Sigma i = 0$

2nd law - voltage law.

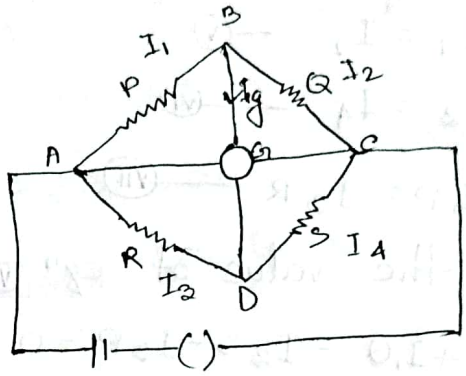
$\Sigma ir = \Sigma \mathcal{E}$

if $\Sigma \mathcal{E}$ is not acting i.e if there is no source to the emf then $\Sigma ir = 0$.

b) Apply the kirchoff's law to a wheatstone bridge and find the conditional for bridge balance.

Ans:
Application of kirchoff's law to wheatstone bridge.





Represent a wheedstone bridge circuit where PQRS are connected to form a closed circuit A cell is connected is between A and C and a galvanometer is connected in between B and D

Apply the kirchiff's 1st law at the junction B

$$I_1 - I_2 - I_g = 0 \quad \text{——(i)}$$

at the junction D

$$I_3 + I_g - I_4 = 0 \quad \text{——(iv)}$$

now Applying the kirchoff's 2nd law in meshes ABDA

$$I_1 P + I_g G - I_3 R = 0 \quad \text{——(iii)}$$

A and ABCDA.

$$I_1 P + I_2 Q - I_4 S - I_3 R = 0 \quad \text{——(iv)}$$

When the galvanometer shows "zero deflection". The point B and D are the same potential and $I_g = 0$

Then put this value of $I_g = 0$ at eqn (I), (II) and (III).

Then we get,

$$I_1 = I_2 \quad \text{—(V)}$$

$$I_3 = I_4 \quad \text{—(VI)}$$

$$I_1 P = I_3 R \quad \text{—(VII)}$$

putting the value of eqn (V), (VI) and (VII) we get,

$$I_1 P + I_1 Q - I_3 S - I_3 R = 0$$

$$\text{or } I_1 (P+Q) - I_3 (S+R) = 0$$

$$\Rightarrow I_1 (P+Q) = I_3 (S+R) \quad \text{—(VIII)}$$

now divide eqn (VIII) by eqn (VII) we get,

$$\frac{I_1 (P+Q)}{I_1 P} = \frac{I_3 (S+R)}{I_3 R}$$

$$\Rightarrow \frac{P+Q}{P} = \frac{S+R}{R}$$

$$\Rightarrow \frac{Q}{P} = \frac{S}{R}$$

$$\text{or } \frac{P}{Q} = \frac{R}{S}$$

and this is the condition for bridge balance

e) show that coulomb's law of electrostatics can be deduced from Gauss' law.

Ans:

Deduction of coloumbs law from Gauss's law.

as we know the first case of Gauss law,

$$E = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q}{r^2}$$

Let put a charge q_0 at the point for which E has been calculated. Then the magnitude of force on q_0 is.

$$F = q_0 E$$

$$F = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q_0}{r^2}$$

That is in a fixed medium. The force between two point charges is directly proportional to the produce of their magnitude and inversely proportional to the squre of their distance and this is the colombs law.

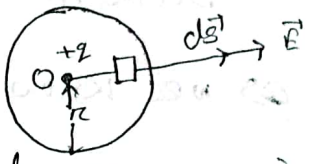
$$f \propto \frac{q q_0}{r^2}$$

$$f = \frac{1}{4\pi\varepsilon_0} \cdot \frac{q q_0}{r^2}$$

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Q. Deduction of coulomb's law from gauss's law:

Let us apply gauss's law to an isolated point charge $+q$ lying in vacuum. the gaussion surface is spherical surface of radius r having its centre at the point charge $\vec{E}$ and $d\vec{s}$ at any point on the surface are directed radially outward and angle between $\vec{E}$ and $d\vec{s}$ is zero.



$$\therefore \vec{E}\cdot d\vec{s} = E.ds\cos 0^\circ$$
$$= Eds$$

Applying gauss's law,

$$\oint \vec{E}\cdot d\vec{s} = \oint E ds = \frac{q}{\epsilon_0}$$

E is constant at all point on the surface.

Thus, $E\oint ds = \frac{q}{\epsilon_0}$ $\quad [\because \oint ds = 4\pi r^2]$

$$\Rightarrow E.(4\pi r^2) = \frac{q}{\epsilon_0}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0}\cdot\frac{q}{r^2}$$

Let us put a second point charge q' on this surface, the force $\vec{F}$ that acts on this charge q' is

$$F = q'E$$
$$= \frac{1}{4\pi\epsilon_0}\cdot\frac{qq'}{r^2}$$

5. a) State and explain Norton's theorem.

Ans: Nortons theorem state that any linear electric circuit containing several energy sources and resistance can be replaced by a single current source in parallel with a single resistor. This equavalent circuit is valid for the external connections to the original circuit, meaning the current. voltage characteristics seek at the terminals of the circuit remain the same.

Nortons equavalent circuit:-

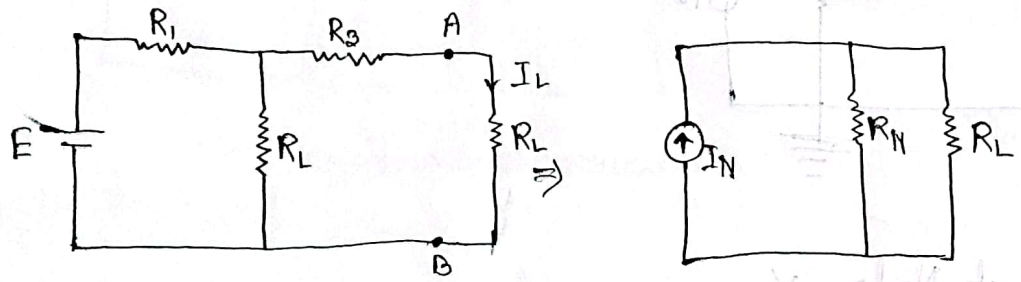
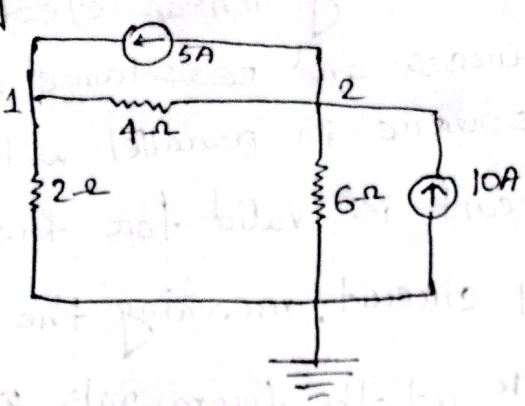


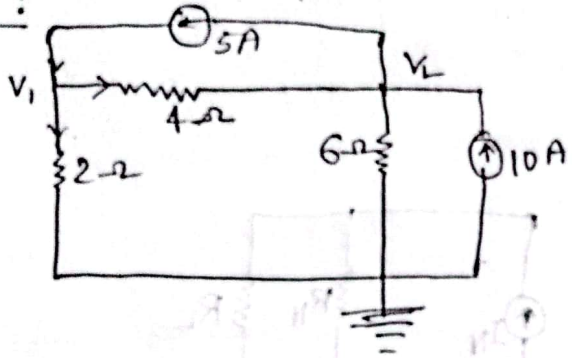
Fig: Norton's Equavalent theorem.

The equavalent circuit consists of a current of a current source I_N in parallel with a resistor 'R_N'. The current source represents the total current flowing out of the original circuit through the load and resistor represent the equavalent resistance of the circuit looking back into the terminals.

b) Calculate the node voltages in the circuit shown in Figure A.



Soln:



Applying KCL at Node V_1,

$$-5 + \frac{V_1}{2} + \frac{V_1 - V_2}{4} = 0$$

$$\Rightarrow \frac{2V_1 + V_1 - V_2}{4} = 5$$

$$\Rightarrow 3V_1 - V_2 = 20 \quad \text{---} (1)$$

Applying KCL at Node V_2,

$$-10 + \frac{V_2}{6} - \frac{V_1 - V_2}{4} = 0$$

$$\Rightarrow \frac{2V_2 - 3V_1 + 3V_2}{12} = 0$$

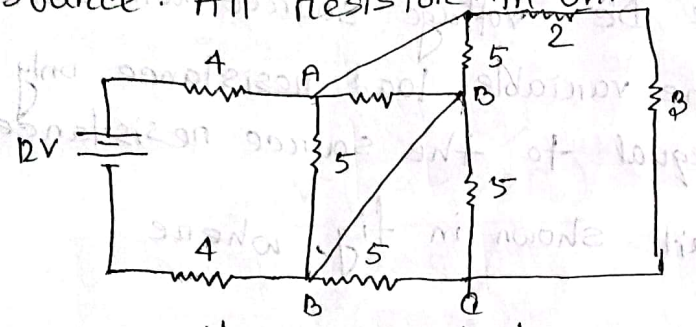
$\Rightarrow -3V_1 + 5V_2 = 120$ ——— (iii)

using calculator in equation (i) and (ii),

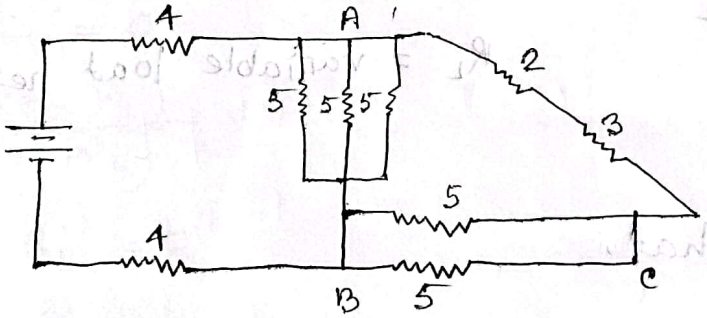
$V_1 = 18.33$

$V_2 = 35$

c) Find the equavalent resistance for circuit shown in figure B. Also calculate the return current of the independent source. All resistor in ohm.



the equavalent circuit.



$$R_T = [\{(5 \parallel 5 \parallel 5) + (5 \parallel 5)\} \parallel (2+3)] + 4 + 4$$

$$= \{(1.67 + 2.5) \parallel 5\} + 8$$

$$= (4.17 \parallel 5) + 8$$

$$= 2.273 + 8$$

$$= 10.273 \text{ (Ans)}$$

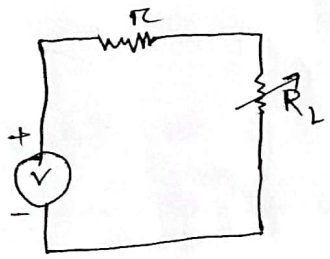
current $I = \frac{V}{R}$

$= \frac{12}{10.273} = 1.17\,A$ (Ans)

6. a) state and prove the maximum power transfer theorem for two terminal network.

Soln: It state that the DC voltage source will deliver maximum power to the variable load resistance only when the load resistance is equal to the source resistance.

proof: consider a circuit shown in fig. where



V = DC voltage

r = internal resistance of DC voltage

R_L = variable load resistance

we know that,

$$I = \frac{V}{r+R_L}$$

the total power connected to the resistive load.

$$P_L = I^2 R_L = \left(\frac{V}{r+R_L}\right)^2 \times R_L$$

P_L can be maximized by adjusting R_L, there highest

Power can be generated when,

$$\frac{dP_L}{dR_L} = 0$$

Now,

$$\frac{dP_L}{dR_L} = \frac{d}{dR_L}\left(\frac{V^2}{(r+R_L)} \times R_L\right)$$

$$= \frac{(r+R_L)^2 \frac{d}{dR_L}(V^2 R_L) - V^2 R_L \frac{d}{dR_L}(r+R_L)^2}{[(r+R_L)^2]^2}$$

$$= \frac{(r+R_L)^2 V^2 - V^2 R_L 2(r+r_L)}{(r+R_L)^4}$$

$$= \frac{V^2(r+R_L-2R_L)}{(r+R_L)^3}$$

$$= \frac{V^2(r-R_L)}{(r+R_L)^3}$$

$$\frac{dP_L}{dR_L} = 0$$

$$\Rightarrow \frac{V^2(r-R_L)}{(r+R_L)^3} = 0$$

$$\therefore r - R_L = 0$$

$$\therefore r = R_L$$

So the highest power transmitted to the load resistance is

$$P_{max} = \frac{V^2}{4r}$$

b) state and explain kirchoff's current and voltage Laws.

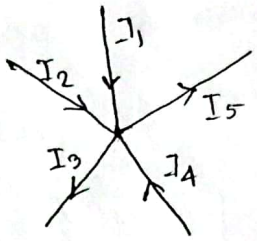
Ans

kirchhoff current Law:-

1. The principle of this law is to conserve the electric charge.

2. The ammount of current flowing into the node is equal to the sum of current flowing out of it.

3. In other word, the algebric sum of all the currents in any given circuit will be equal to zero.



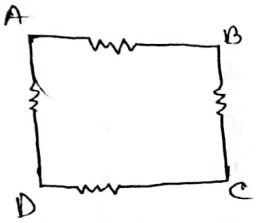
$$I_1 + I_2 + I_4 - I_3 - I_5 = 0$$

kirchhoff voltage law:-

1. The principle of the law is to conserve energy.

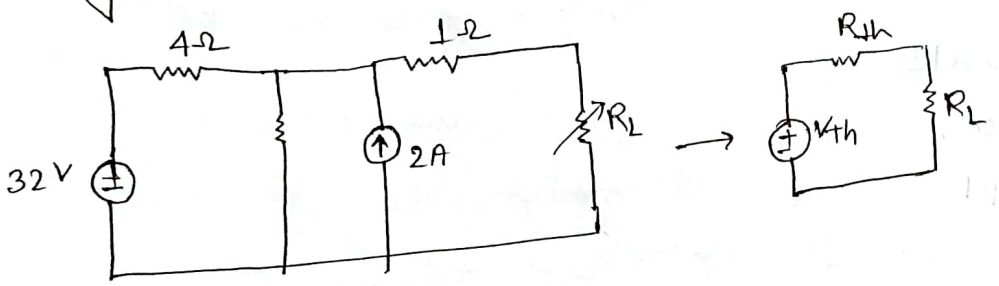
2. The law states that the sum of voltages in a closed loop is zero.

3. The total amount of energy gained is equal to the energy lost per unit charge

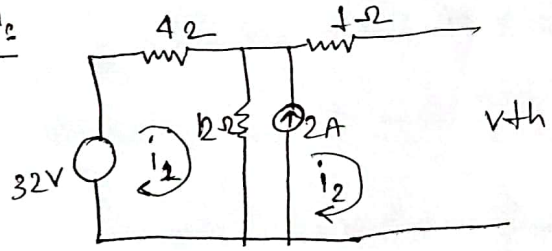


$V_{AB} + V_{BC} + V_{CD} + V_{DA} = 0$

c. Find the thevenin equivalent circuit of the circuit shown in figure. Then find the current through $R_L = 6\Omega$



Sol^n:



Applying KVL at loop i_1

$-32 + 4i_1 + 12(i_1 - i_2) = 0$

$\Rightarrow -32 + 4i_1 + 12i_1 - 12i_2 = 0$

$\Rightarrow 16i_1 - 12i_2 = 32$

$\Rightarrow 4i_1 - 3i_2 = 8$

$\Rightarrow 4i_1 - 3(-2) = 8$

$\therefore i_2 = 2$

$\Rightarrow 4i_1 = 8 - 6$

$\Rightarrow i_1 = \frac{2}{4} = 0.5$

$I_{i_2} = i_1 - i_2$

$= 0.5 - (-2)$

$= 2.5$

$V_{th} = V_{12} = i_{12} \times 12$

$= 2.5 \times 12$

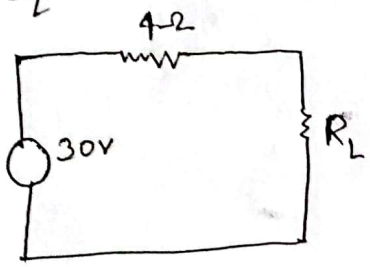
$= 30\ V$

$R_{th} = (4 \| 12) + 1$

$= 3 + 1$

$= 4\ \Omega$

∴ The equavalent circuit is



when $R_L = 6$

current $I = \frac{V}{4+6}$

$= \frac{30}{10}$

$= 3A$ (Ans)

7. a) Define instantaneous voltage and RMS voltage.

Ans:

Instantaneous voltage refers to the voltage of an electrical signal at a specific point in time. In alternating current (AC) circuit, the voltage is constantly changing direction and magnitude, following a sinusoidal or other waveforms. The instantaneous voltage is the voltage at any given moment along this waveform.

It is represented mathematically as $v(t)$, where t is the time.

$$v(t) = V_m \sin(\omega t + \phi)$$

where V_m is the maximum voltage.

ω is angular frequency

ϕ is the phase angle.

RMS voltage:-

Root Mean Square voltage is a way to represent the effective or equivalent voltage of an AC waveform.

For a sinusoidal wave form, the RMS voltage is the square root of the average of the square of all instantaneous

voltage values over one complete cycle.

Mathematically for a sinosoidal wave form ~~$v(t) = V_m \sin(\omega t$~~

$$v(t) = V_m \sin(\omega t + \phi)$$

∴ the RMS voltage,

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v(t)^2 \, dt}$$

where T is the period of the wave form.

b) Draw the timing and vector diagram for pure resistive. inductive and capacitive load. what is reactance and impedance of AC circuit?

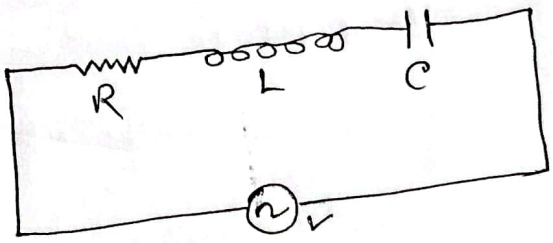
c) Find Impedance, total current, voltage drop across each elements, draw voltage vector, power absorbed by resistance power factor from,

$R = 25\Omega$

$L = 10mH$

$C = 250\mu f$

$V = 230V, 60Hz$



Soln:

we know,

$$X_L = 2\pi fL$$

inductive Reactances:

$$= 2 \times 3.1416 \times 50 \times 10 \times 10^{-3}$$

$$= 3.14\Omega$$

Capacitive Reactance $X_C = \frac{1}{2\pi fC}$

$$= \frac{1}{2 \times 3.1416 \times 50 \times 0.00025}$$

$$= 12.74\Omega$$

Total impedance $Z = R + j(X_L - X_C)$

$$= 25 + j(3.14 - 12.74)$$

$$= 26.78 \angle -21.0$$

Total current $I = \frac{V}{Z}$

$= \frac{220\angle 0^\circ}{26.78\angle -21^\circ}$

$= 8.21\angle 21^\circ$

Calculating voltage drop

Across resistor $V_R = I \times R$

$= (8.21\angle 21^\circ) \times .25$

$= 205.25\angle 21^\circ$

Across inductor $V_L = I \times X_L$

$= (8.21\angle 21^\circ) \times 3.14$

$= 25.78$

Across the capacitor $V_C = I \times X_C$

$= (8.21\angle 21^\circ) \times 12.74$

$= 104.59$

Calculating power Asorved by Resistance,

$P_R = I^2R$

$= (8.21)^2 \times 25$

$= 1685.1023 w$

power factor $= \cos(\theta)$

$= \cos(21)$

$= 0.93$

The voltage vector diagram-

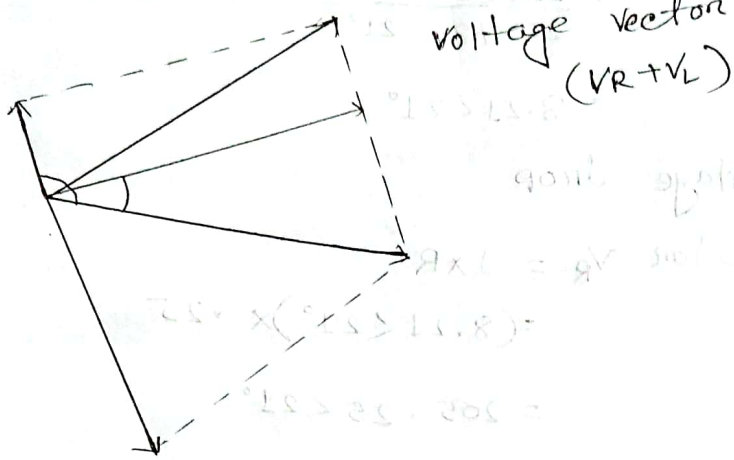


Fig: vol

8. a) write down the difference between steady and transient current.

	Steady current	transient current
Defination	i) constant flow of electric charge	temporary or changing flows
time dependency	time invarient	time varying
Behaviour over time	stable and uncharging	fluctuate or change over time
example	Dc circuits, steady, state AC.	changing or discharging events.
mathematical model	connect remain constant	current changes with time.

b) Define time constant for suitable example.

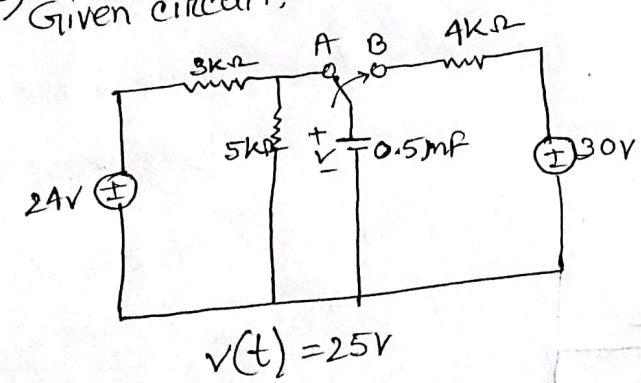
Sol^{n}: The time constant often denoted as τ (tau). The time required for a current turned into the circuit under a steady electromotive force to reach to on 0.632 of its final strength.

If time constant τ, resistance R and capacitence C.

Then $\tau = R.C$

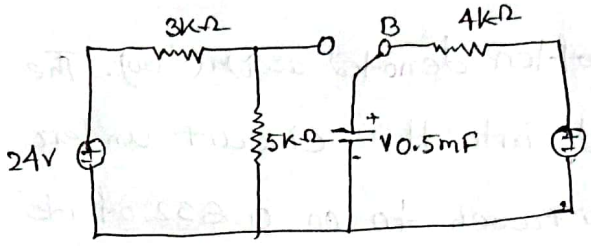
For example If you have a circuit with $R = 1K\Omega$ and $C = 1\mu f$ the time constant could be 1ms. This implies that it takes approximately 1ms for the voltage across the capacitor to reach around 63.2%. of its final value in response to sudden change in voltage.

c) Given circuit,



At t=0 the switch moves to B and that time the circuit

is source circuit, the circuit is



we know the formula.

$$v(t) = v(\infty) + [v(0) - v(\infty)]\, e^{-t/\tau} \quad \text{— (1)}$$

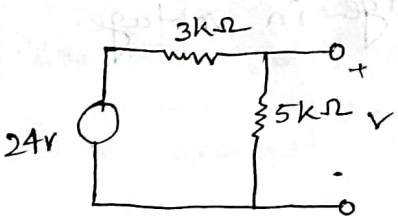
first we have to calculate

$v(\infty)$ = when $t > 0$

$v(0)$ = " $t < 0$

τ = " $t > 0$

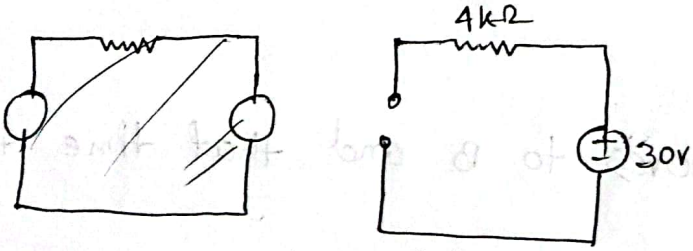
when $t < 0$, then circuit is.



$$v(0) = \frac{5}{3+5} \times 24$$

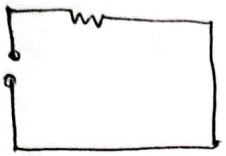
$$v(0) = 15V$$

when $t > 0$ the circuit is



$v(\infty) = 30V$

when $t>0$ then R_{eq}



$R_{eq} = 4k\Omega$

we know

$\tau = R_{eq} C$

$= (4 \times 10^3) \times (0.5 \times 10^{-3})$

$\tau = 2s$

we obtained

$v(0) = 15V$

$v(\infty) = 36V$

$\tau = 2s$

put this value in eqn (i)

$$v(t) = v(\infty) + [v(0) - v(\infty)] e^{-t/\tau}$$

$$\Rightarrow 25 = 30 + [15-30] e^{-t/2}$$

$$\Rightarrow 25 = 30 + (-15) e^{-t/2}$$

$$\Rightarrow 30 + (-15) e^{-t/2} = 25$$

$$\Rightarrow 30 - 15 e^{-t/2} = 25$$

$$e^{-t/2} = 5/15$$

$$\therefore -t/2 = \ln(5/15)$$

$$\Rightarrow -t/2 = -1.098$$

$$\Rightarrow t = 1.098 \times 2$$

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